

Lab 4: Bias-Variance Tradeoff and Regularization

Dataset: [California Housing Dataset](#)

Use the **California Housing Dataset** and investigate how polynomial model complexity and L2 regularization affect regression performance.

1. Load and Explore the Dataset

- Load the California Housing dataset.
- Select one numerical feature as X and the target variable as y.
- Create a scatter plot to visualize their relationship.

2. Polynomial Regression

- Create polynomial features using at least three different degrees, such as 1, 2, and 5.
- Train a Linear Regression model for each degree.
- Calculate and compare the Training MSE.

3. L2 Regularization

- Choose a higher polynomial degree, such as degree 5 or 10.
- Train the model with different values of λ , for example: 0, 0.1, 1, 10, and 100.
- Record the Training MSE for each value.

4. Visualization

- Plot the predictions of the polynomial model with and without regularization on the same graph.
- Observe how changing λ affects the model.

5. Ridge Regression

- Use sklearn's **Ridge** model with the same polynomial degree and regularization strength.
- Calculate its MSE and compare it with your manually implemented L2 regularization.

Submission: Submit your notebook with the code, relevant plots, MSE results, and your conclusion.